KiranaCrew

VOICE-FIRST AI WORKFORCE FOR INDIAN KIRANA STORES

A shopkeeper speaks. A coordinated team of AI agents gets the work done.

GNANI AI HACKATHON

PRISMA v2.5
Speech-to-text

TIMBRE v3.0
Text-to-speech

EVON | future reasoning vision (not integrated)

01 THE PROBLEM

Small stores run on big operational complexity. India has an estimated **12 million kirana stores**. Owners manage inventory, customer udhaar, supplier orders, expenses and billing—often with paper registers, WhatsApp and memory. Many are more comfortable speaking Hindi or Hinglish than navigating English-first software.

02 THE SOLUTION

KiranaCrew turns everyday speech into coordinated action. Owners speak naturally in Hindi, English or Hinglish. Alice, the manager agent, interprets intent and routes work to specialist agents. They retrieve live business information or prepare actions, while sensitive billing stays behind owner approval.

03 HOW IT WORKS | FROM SPOKEN REQUEST TO VERIFIED ACTION

1 · SPEAK	2 · TRANSCRIBE	3 · ROUTE	4 · EXECUTE	5 · RESPOND
Owner asks naturally in Hindi, English or Hinglish.	Gnani Prisma converts speech into text; Hinglish detection helps select language mode.	Alice selects one or more specialist agents for the request.	Agents use ERPNext through MCP to read records or prepare store actions.	Gnani Timbre streams a natural spoken answer back.

04 THE 7-SPECIALIST AI WORKFORCE

ALICE · MANAGER	Routes requests, coordinates agents and oversees task execution.	**BOB · INVENTORY**	Checks stock levels and flags low-stock items.
CAROL · SALES	Tracks customer udhaar and credit balances.	**DAVE · PROCUREMENT**	Compares suppliers and prepares purchase orders.
ERIN · FINANCE	Summarizes cash flow and expenses.	**GRACE · REPORTS**	Creates daily and weekly business reports.
FRANK · BILLING	Creates invoices; owner PIN approval is required before posting.	**COORDINATION**	Related specialists can work together—for example, finance + inventory for a combined check.

05 GNANI IS THE VOICE LAYER

Prisma v2.5 STT: live microphone speech recognition configured for Roman-script Hinglish (**hl**), Hindi (**hi-IN**) and English (**en-IN**). **Timbre v3.0 TTS:** WebSocket-streamed spoken replies in Indian English/Hindi, with 13 available voices and Avinash as default. A custom **HINGLISH_WORDS** detector helps select language mode. Gemini is configured as backup for Gnani 5xx errors or timeouts.

06 BUILT FOR REAL STORE OPERATIONS

ERPNext + MCP: agents connect to stock, invoice and supplier records. **Human approval:** billing cannot post until the owner enters a PIN. **Latency:** parallel ERP reads using **Promise.all** reduce measured workflow time by approximately **900 ms**. **Testing:** 73 passing tests across the voice pipeline.

07 A REAL KIRANA MOMENT

“Kal kitna udhaar tha Ramesh ka?”

KiranaCrew transcribes the question with Prisma, queries the relevant customer-credit records in ERPNext, and speaks the answer using Timbre. The target experience is a response in under three seconds—with no typing and no menu navigation.

08 IMPACT & ENGINEERING EVIDENCE

12M	7	73	~900 ms
Kirana stores in the target market	specialist agents in one workforce	passing tests across the voice pipeline	reduction from parallel ERP reads

WHY KIRANACREW IS DIFFERENT

Not another chatbot or dashboard: KiranaCrew combines a voice-native interface, specialist agents, live operational data and an explicit human approval boundary for billing.

THE VISION

Help independent shop owners make informed decisions and complete routine work in the language they already use—while keeping them in control of consequential actions.

LIVE DEMO
youtube.com/watch?v=ID-dp83yWhI

GITHUB REPOSITORY
github.com/rajatnagda45/KiranaCrew

TECH STACK
Gnani AI · ERPNext · MCP